ST MARY'S DSG, KLOOF

GRADE 10 **MATHEMATICS - PAPER 1** **NOVEMBER 2018**

TIME: 2 HOURS **TOTAL MARKS: 135**

NAME: Memo

MATHS TEACHER: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This paper consists of 18 pages and 16 questions. Please check that your question paper is complete.
2. There are 5 sections (A – E). Write your name and Mathematics teacher's name on each section. Hand each section in separately.
3. Answers, where necessary, must be rounded off to two decimal digits.
4. Write neatly and show your working.
5. An approved non-programmable calculator may be used unless otherwise stated.

FORMULAE YOU MAY NEED

$A = P(1 + in)$ $\quad A = P(1+i)^n$ $\quad A = P(1-in)$ $\quad A = P(1-i)^n$

$P(A) = \frac{n(A)}{n(S)}$ $\quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$P(A) + P(A') = 1$

$y = mx + c$ $\quad m = \frac{y_2 - y_1}{x_2 - x_1}$

SECTION A [29 Marks]

QUESTION 1

Factorise the following completely:

a) $a^3 + 27$ (3)

$= (a+3)(a^2 - 3a + 9)$

b) $2b^2 - 10b + 12$ (3)

$= 2(b^2 - 5b + 6)$

$= 2(b-3)(b-2)$

c) $e^2(2d - 3e) + 4(3e - 2d)$ (3)

$= (e^2 - 4)(2d - 3e)$

$= (e-2)(e+2)(2d-3e)$

[9]

QUESTION 2

Simplify fully, leaving answers with positive exponents.

a) $(2a - 1)(a^2 - 2a - 1)$ (2)

$= 2a^3 - 4a^2 - 2a - a^2 + 2a + 1$

$= 2a^3 - 5a^2 + 1$

b) $\frac{x^2+2x}{x^2-x-6} \div \frac{1}{x-3}$ (4)

$= \frac{x(x+2)}{(x-3)(x+2)} \times \frac{x-3}{1}$

$= x$

c) $\frac{2}{x} - \frac{x-2}{x+1}$ (3)

$= \frac{2(x+1)}{x(x+1)} - \frac{x(x-2)}{x(x+1)}$

$= \frac{2x+2-x^2+2x}{(x)(x+1)}$

$= \frac{4x+2-x^2}{x(x+1)}$

d) $\frac{2^{n-3}.3^{n+1}}{6^n.4^{-2}}$ (4)

$= \frac{2^{n-3}.3^{n+1}}{2^n.3^n.2^{-4}}$

$= 2^{n-3-n+4}.3^{n+1-n}$

$= 2.3$

$= 6$

e) $\frac{2^{x+2}-2^{x+1}}{2^x+2^{x+2}}$ (4)

$= \frac{2^x(2^2-2)}{2^x(1+2^2)}$

$= \frac{2}{5}$

[17]

QUESTION 3

A second-hand farm tractor worth R60 000 has a limited useful life of 5 years and depreciates at 20% p.a. on a reducing-balance basis. Determine the value of the tractor at the end of the 5 year period.

$A = 60000(1 - 20\%)^5$

$= R19660,80$

[3]

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SECTION B [26 Marks]

QUESTION 4

4.1 Solve for x:

a) $2^{x+1} = 9 - 2^0$ (3)

$2^{x+1} = 8$

$2^{x+1} = 2^3$

$x = 2$

b) $(x-7)(x-5) = 3$ (4)

$x^2 - 12x + 35 = 3$

$x^2 - 12x + 32 = 0$

$(x-4)(x-8) = 0$

$x = 4$ or $x = 8$

4.2 Solve for x and represent your answer on a number line.

$\frac{-x+3}{2} < 1$ for $x \in R$ (3)

$-x + 3 < 2$

$-x < -1$

$x > 1$



4.3. Solve for x and y in the following set of equations:

$y - 3x = -4$ ① and $5x + 2y = 3$ ② (4)

$y = 3x - 4$ Sub ① into ②

Sub x into ① $5x + 2(3x - 4) = 3$

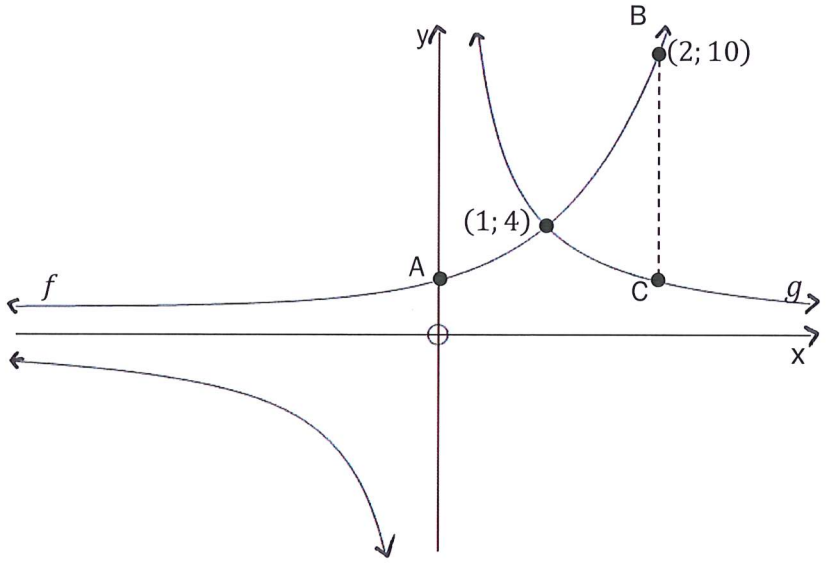
$y = 3(1) - 4$ $11x = 11$

$= -1$ $x = 1$

[14]

QUESTION 5

You are given the following sketch of the functions: $f(x) = 3^x + 1$ and $g(x) = \frac{k}{x}$
f and g intersect at $(1; 4)$ and B is the point $(2; 10)$



a) Determine the coordinates of A. (2)

(0; 2)

b) Determine the value of k. (1)

$k = 4$

c) Give the equation for the asymptote of the graph $y = 3^x + 1$ (1)

$y = 1$

d) Write down the range of f. (2)

$y \in (1; \infty)$

e) For which values of x is $g(x)$ a decreasing function? (3)

$x \in (-\infty; 0)$ and $(0; \infty)$

g) Determine the distance of BC. (3)

$y = \frac{4}{2}$ $\quad BC = 10 - 2$

$y = 2$ $\quad BC = 8$

[12]

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SECTION C [25 Marks]

QUESTION 6

When you were on The Journey in Grade 9, you took frequent breaks to drink water, eat snacks, reapply sunscreen, and appreciate the views.

You noticed that your group took slightly longer breaks every time you stopped, according to the following pattern:

4 min; 9 min; 14 min ...

a) Write down an expression for the nth term in the sequence above. (2)

$T_n = 5n - 1$

b) Calculate the length of your team's 12th break, if the pattern was to continue in the same way. (2)

$T_{12} = 5(12) - 1$

$= 59$ min

c) If the pattern stays the same, will your team ever take a break which is exactly 2 hours long? (2)

$120 = 5n - 1$

$121 = 5n$

$24,2 = n$

$\therefore$ no

[6]

QUESTION 7

Inside Kayleigh's pencil case there are 16 blue pens, 4 red pens and 3 erasers.

a) If Kayleigh puts her hand into her pencil case and takes out an item of stationery, what is the probability that it will be:

i. a red pen. (1)

$\frac{4}{23}$

ii. a blue pen or an eraser (2)

$\frac{19}{23}$

b) Emma selects 2 items of stationery out of Kayleigh's pencil case which she picks out one at a time. Assuming she does not replace the item of stationery, what is the probability that she will select a red pen followed by an eraser? (3)

$\frac{4}{23} \times \frac{3}{22} = \frac{6}{253}$

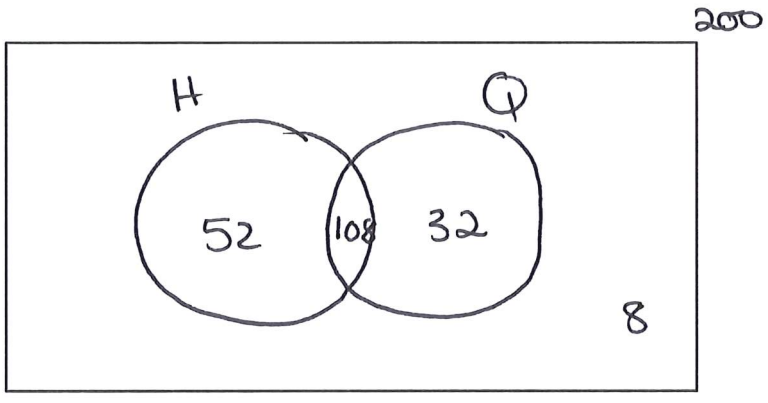
[6]

QUESTION 8

In a survey of a group of 200 teenage girls it was found that:

160 liked Hip-hop (H) music
140 liked Qwaito (Q) music
8 did not like either type of music

a) Draw a Venn diagram illustrating the given information. (5)



b) Use your Venn diagram in QUESTION 8a) to calculate the probability that a teenager, randomly chosen,

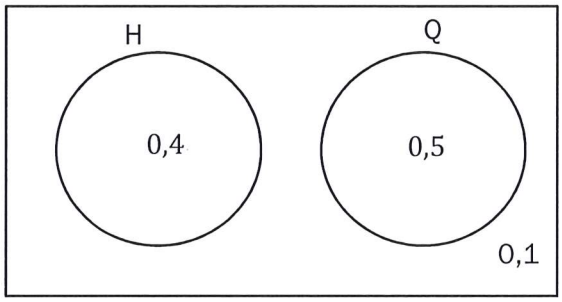
i. Likes Qwaito or Hip-Hop music (2)

$\frac{192}{200}$

ii. Likes Qwaito but not Hip-Hop (2)

$\frac{32}{200}$

c) A group of boys is also surveyed and the probability is shown in the Venn diagram below:



Are the events liking Hip-Hop and not liking Hip-Hop complementary? Use a probability formula to justify your answer. (2)

$P(H) + P(H')$

$= 0,4 + 0,6$ ∴ yes

$= 1$

[11]

QUESTION 9

Given $A = \sqrt{4x - 3}$

Determine for each of the following ONE value of x that satisfies the condition:

a) A is irrational (1)

Any value that doesn't give a p.s or < 0

b) A is non-real (1)

Any value $< \frac{3}{4}$

[2]

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SECTION D [27]

QUESTION 10

The diagram below shows figures 1, 2 and 3 being made up of different amounts of coloured blocks. For example, figure 1 comprises of 5 blocks.

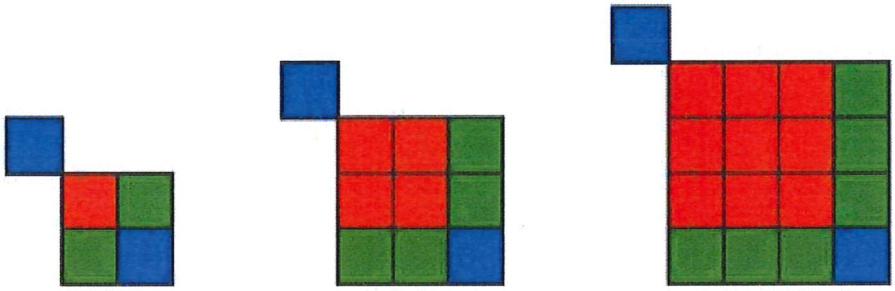


Figure 1 Figure 2 Figure 3

a) How many red blocks will there be in figure 11? (1)

121

b) Determine a formula for the number of blocks for figure n. (4)

$T_n = n^2 + 2n + 2$

c) Showing working, determine which figure would be made up of 325 blocks. (3)

$325 = n^2 + 2n + 2$

$0 = n^2 + 2n - 323$

$0 = (n - 17)(n + 19)$

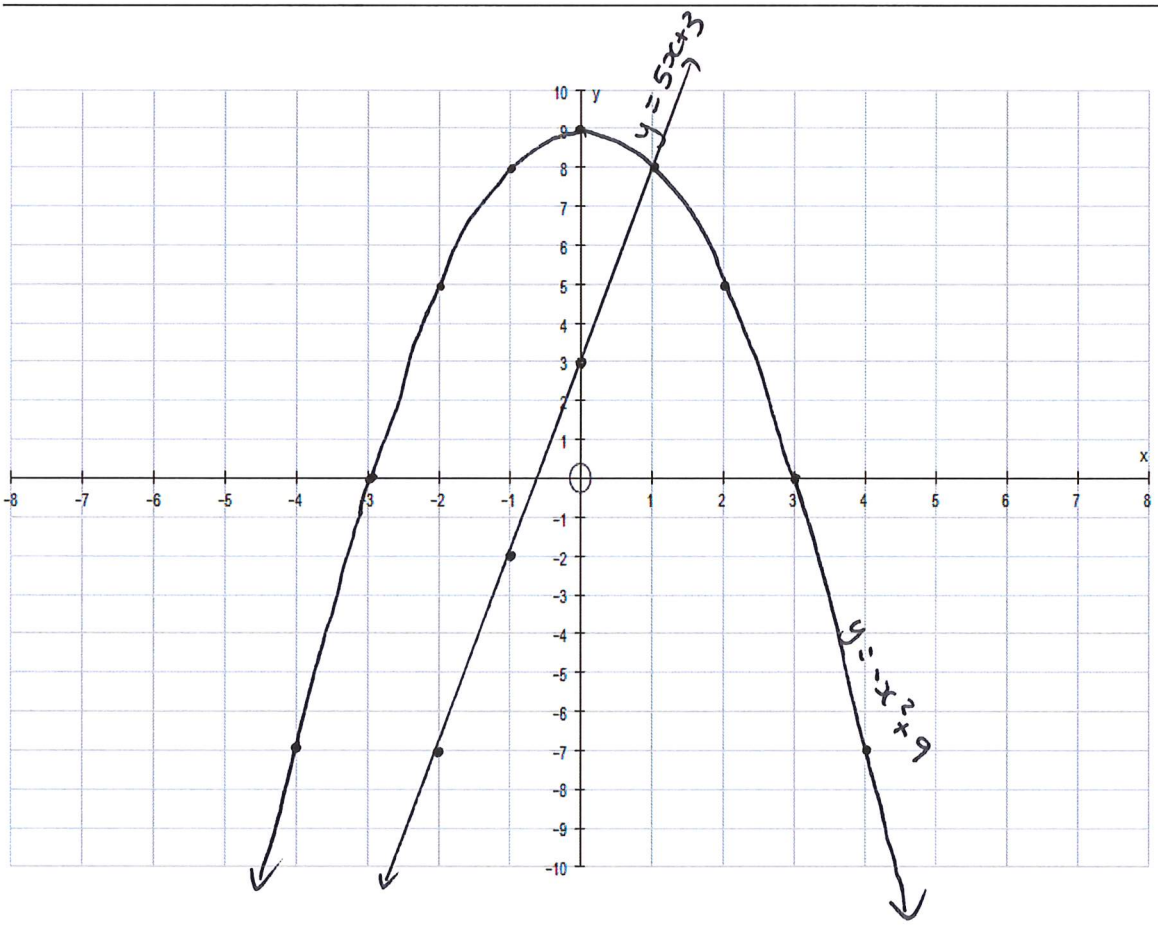
$\therefore$ figure 17

[8]

QUESTION 11

You are given the functions: $f(x) = -x^2 + 9$
$g(x) = 5x + 3$

a) Sketch the graphs of f and g on the same set of axes. (5)



b) Write down the equation of the axes of symmetry of $f(x)$. (1)

$y = 0$

c) For which value(s) of x is $f(x) > 0$? (2)

$x \in (-3; 3)$

d) Give the equation of $p(x)$, which is a reflexion of f in the x-axis. (2)

$p(x) = x^2 - 9$

e) Determine the formula of the straight line that passes through (10;-4) and is perpendicular to $g(x)$. (3)

$y = -\frac{1}{5}x + c$

$-4 = -\frac{1}{5}(10) + c$

$-4 + 2 = c$

$y = -\frac{1}{5}x - 2$

f) Give the equation of $h(x)$, if h is created by shifting the graph of f down by 3 units. (1)

$h(x) = -x^2 + 6$

g) Determine algebraically the point(s) at which $f(x)$ and $g(x)$ intersect. (5)

$-x^2 + 9 = 5x + 3$

$0 = x^2 + 5x - 6$

$0 = (x - 1)(x + 6)$

$x = 1$ and $x = -6$

$\therefore y = 5(1) + 3$ $\quad y = 5(-6) + 3$

$= 8$ $\quad = -27$

$(1; 8)$ $\quad (-6; -27)$ **[19]**

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SECTION E [28 Marks]

QUESTION 12

A student plans to buy a car in 6 years' time so she starts saving by depositing R20 000 into a savings account. After 2 years she deposits a further R35 000. The interest on her money is compounded at 11% per annum for the first 2 years and then at 10,5% per annum for the remaining years of the investment.
How much will she have saved at the end of the 6 years?

$20000(1+11\%)^2(1+10,5\%)^4$

$+ \quad 35000(1+10,5\%)^4$

$= \quad R\ 88920,38$

[5]

QUESTION 13

If I travel at $120km/h$ instead of $100km/h$, I will save 36 minutes on a trip. What distance would I have travelled?

Let distance be x

$$\frac{x}{100} - \frac{x}{120} = \frac{36}{60}$$

$$\frac{6x - 5x}{600} = \frac{360}{600}$$

$x = 360$ km

[5]

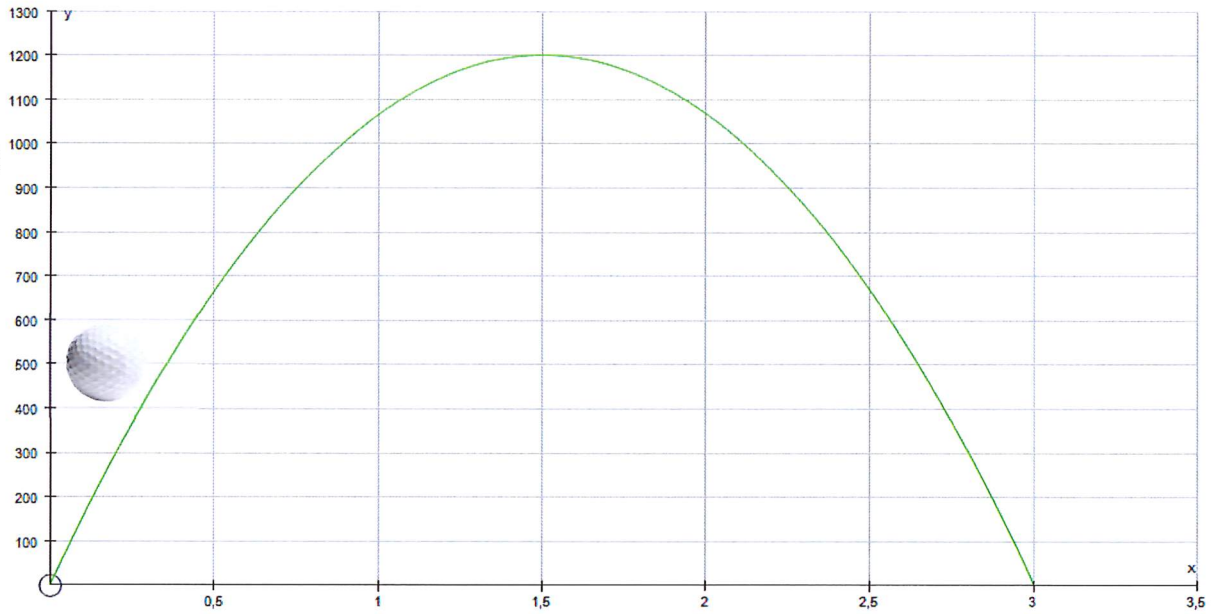
QUESTION 14

On the final day of the 2018 British Open Golf Tournament, Francesco Molinari chipped his golf ball off the fairway onto the green to win.

The path that the golf ball followed fits the equation: $h(t) = -533t^2 + 1600t$, where h is the height in centimetres and t is the time in seconds from the time the ball is hit.

The domain is $t \in [0; 3]$

Use the graph to answer the questions that follow



a) At what time (in seconds) was the height zero? What does this mean for the golf ball? (2)

0 and 3 Seconds

Ball is on the ground

b) How long was the golf ball in the air for? (1)

3 Seconds

c) How long after being hit does the golf ball reach its maximum height? (1)

1,5 Seconds

d) For approximately how long is the ball above 900 cm? (2)

1,5 Seconds

e) Explain why t is restricted to positive values in this situation. (1)

Time cannot be negative

[7]

QUESTION 15

A disease is killing off a population of fish in a dam at a rate of 10% every 24 hours.

The function $p(x) = A(1-i)^t$ is given, where A is the initial population, i is the rate of decrease and t is the number of days that have passed.

The following data is collected by researchers:

Days after disease is identified	0	1	2	3
Fish Population	A	4500	4050	3645

a) Determine A the initial population of fish in the dam (3)

$4500 = A(1-10\%)$

$5000 = A$

b) Between which two consecutive days will the population of the fish be halved? (3)

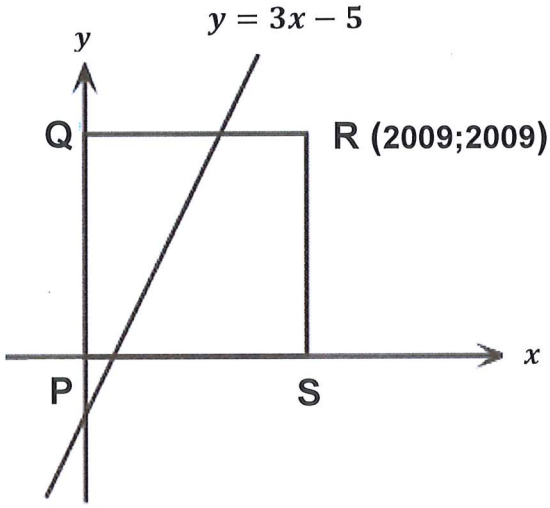
$2500 = 5000(1-10\%)^n$

Between 6 and 7 days

[6]

QUESTION 16

A *lattice point* is a point where both coordinates are integers.
For example $(1;4)$ is a lattice point but $\left(\frac{3}{4};4\right)$ is not.
The line $y = 3x - 5$ passes through the square $PQRS$ as shown in the diagram.
If the coordinates of R are $(2009;2009)$ how many lattice points on the line lie inside the square?



$2009 = 3x - 5$

$2014 = 3x$

$671\frac{1}{3} = x$

$\therefore$ 671 is max x-value.

x-int: $0 = 3x - 5$ **[5]**

$\frac{5}{3} = x$

$\therefore$ next x value must be 2

$\therefore 671 - 1 = 670$